# D. KARTHIKA

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## EXECUTIVE SUMMARY

Software Engineering student with experience in Full-Stack Development and AI/ML projects.Passionate about building scalable applications, intelligent systems, and data-driven solutions.

## EDUCATION

**Vellore Institute of Technology, Vellore** *July 2021 - August 2026*
M. Tech (Integrated) – Software Engineering CGPA: 7.60/10

**SRV Girls Hr Sec School , Rasipuram** *May 2021*
TNBHSE (12th Board) Percentage: 80.89

**EBENEZER Matric Hr Sec School,Yelagiri Hills,Vellore** *May 2019*
TNBSE (10th Board) Percentage: 71.8

## TECHNICAL STRENGTHS

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|---|---|
| **Programming Languages** | **Python** |
| **Databases** | **SQL,MySQL,PostgreSQL** |
| **Full-Stack Technologies** | **React.js, HTML5, CSS3, JavaScript, Node.js, Express.js, FastAPI, REST APIs** |
| **AI/ML** | **Pandas, NumPy, Matplotlib, Seaborn,Scikit-learn, Tensorflow, Keras, LSTM,XGBoost, LightGBM** |
| **Tools & Platforms** | **Git, GitHub, VS Code** |
| **Software Engineering** | **Agile, SDLC, Software Testing, Data Structures Algorithms, opps.** |

## EXPERIENCE

**Coptercode** june - july 2026
*Full Stack Development Intern*

· Contributing to the development of a real-time Chat Application System using React.js, Node.js, Express.js, FastAPI, and PostgreSQL. Implementing AI-powered features including spam message detection and a Knowledge Tracker module for personalized course recommendations.

## PROJECTS

**AI-Based Fleet Maintenance**

- Designed and developed an AI-based aircraft engine predictive maintenance system using LSTM, XGBoost, LightGBM, and SHAP to predict Remaining Useful Life (RUL), analyze sensor degradation patterns, provide explainable predictions, and visualize engine health through an interactive dashboard.

**Chat Application System**

- Built a full-stack Chat Application System with real-time messaging, AI-powered spam detection, and personalized learning recommendations using React.js, Node.js, FastAPI, PostgreSQL,YouTube API, and SerpAPI